# Professional Practice PP Spring-25

## Short Questions

### 1. Explain the philosophical view of ethics?

The philosophical view of ethics means the basic thinking about what is right and wrong. Ethics helps people decide how they should behave in personal and professional life.

- Ethics is based on moral values like honesty, fairness, and responsibility
- It teaches people to do the right thing even when it is difficult
- It focuses on good behavior towards others
- It helps in making correct decisions in difficult situations

In professional life, ethics guide people to work honestly, respect others, and avoid harmful actions. The philosophical view of ethics helps create a responsible and fair society.

### 2. Shortly explain the types of software houses in Pakistan?

In Pakistan, there are different types of software houses.

**1. Product-Based Software Houses**

- They develop their own software products
- Sell software in the market
- Focus on innovation

Example: Mobile apps, management systems

**2. Service-Based Software Houses**

- Provide services to clients
- Work on client requirements
- Earn through projects

Example: Web development, freelancing projects

**3. Outsourcing Software Houses**

- Work for foreign clients
- Complete international projects
- Earn foreign revenue

**4. Startup Software Houses**

- Small companies with new ideas
- Focus on innovation and growth

### 3. Explain the typical structure of a software house?

A software house develops software and provides IT services. It is divided into different departments to work efficiently.

**1. Management Team**

Plans projects, manages resources, and assigns tasks.

**2. Development Team**

Programmers write code and build software.

**3. Quality Assurance (QA) Team**

Tests software for bugs and ensures quality.

**4. Design Team**

Designs user interface (UI/UX) for easy use.

**5. Marketing and Sales Team**

Promotes products and finds clients.

**6. Support Team**
Helps customers and solves their problems.
All teams work together to deliver high-quality software efficiently.

## 4. What are the types of "computer contracts"? Explain any two?

Computer contracts are agreements between parties related to software or IT services.
**Types:**

- Software development contract
- Maintenance contract
- Licensing contract
- Outsourcing contract

**(i) Software Development Contract**
This contract is made between a client and a developer.

- Developer creates software according to requirements
- Client pays for the work
- Includes timeline and cost

Example:
A company hires a developer to build a website.

**(ii) Licensing Contract**
This contract gives permission to use software.

- User can use software legally
- Ownership remains with the company
- Includes terms and conditions

Example:
Buying a licensed copy of software.

## 5. What are the main "health and safety measures" at work?

Health and safety measures are rules to protect workers from harm.
**1. Clean and Safe Environment**
Workplace should be clean and organized.
**2. Proper Lighting**
Good lighting reduces eye strain.
**3. Ergonomic Furniture**
Chairs and desks should be comfortable.
**4. Electrical Safety**
Safe use of computers and wires.
**5. Fire Safety**
Fire alarms and extinguishers should be available.
**6. Breaks and Rest**
Employees should take breaks to avoid stress.
These measures help keep employees safe and healthy.

## 6. Explain shortly 'hacking'?

Hacking means accessing a computer system without permission. Hackers use different techniques to break into systems, such as guessing passwords, using viruses, or finding weaknesses in software.

- Hackers try to steal data or damage systems
- It is illegal and harmful

- Can cause financial and data loss

**Types:**
- Ethical hacking (legal)
- Malicious hacking (illegal)

Hacking is a serious issue and strong security is needed to prevent it.

## Long Questions

### 1. Give an overview of the IEEE Code of Ethics in details?

The Institute of Electrical and Electronics Engineers Code of Ethics is a guide for engineers and professionals to behave in a responsible and ethical way. It ensures that all work is done honestly, safely, and for the benefit of society.

**1.Honesty and Integrity**
- Professionals must always tell the truth
- Provide correct and accurate information in reports and projects
- Avoid cheating, lying, or misleading others

Example:
An engineer reports real test results of a software even if it is not perfect.

**2. Responsibility to Society**
- Engineers must work for the benefit of the public
- Avoid actions that may harm people or the environment
- Focus on improving the quality of life for everyone

Example:
Designing safe electrical systems to prevent accidents.

**3. Avoid Conflict of Interest**
- Professionals should not use their position for personal gain
- Decisions should be fair and unbiased

Example:
A software engineer avoids giving a contract to a relative company if it is not best for the project.

**4. Professional Competence**
- Professionals must maintain their skills
- Keep learning new technology and methods
- Deliver high-quality work

Example:
Regularly attending workshops and learning new programming languages.

**5. Respect for Others**
- Treat colleagues, clients, and users with respect
- Avoid discrimination and harassment
- Promote teamwork and cooperation

Example:
Respecting the ideas of junior team members in a project.

### 6. Honesty in Research and Reports

- Avoid plagiarism
- Give credit to original authors
- Present data and findings truthfully

Example:
Citing sources in a research paper about AI systems.

The IEEE Code of Ethics helps engineers work honestly, responsibly, and fairly, which protects society and improves professional standards.

## 2. Give an overview of ACM/IEEE Software Engineering Code of Ethics and Professional Practice in details?

The ACM/IEEE Software Engineering Code of Ethics is a detailed guide for software engineers. It ensures software is developed ethically, safely, and responsibly.

### Public Interest

- Software should benefit the public and avoid harm
- Ensure safety, reliability, and privacy

Example:
A hospital software system should protect patient data and not crash during emergencies.

### Client and Employer Responsibilities

- Work honestly for clients and employers
- Protect confidential information
- Fulfill contractual obligations

Example:
Not sharing a client's project details with competitors.

### Product Quality

- Deliver high-quality software
- Test software properly before release
- Avoid releasing unsafe or unreliable software

Example:
A banking app must not have security flaws.

### Professional Judgment

- Make fair, independent decisions
- Avoid taking instructions that are unethical or illegal

Example:
Refusing to implement spyware in a product.

### Management Responsibility

- Managers should encourage ethical behavior
- Promote fairness and teamwork
- Provide support and guidance to employees

Example:
A project manager ensures no one is overworked and deadlines are fair.

**Responsibility to Profession**
- Maintain and improve the reputation of software engineering
- Follow ethical standards and professional practices

Example:
Promoting proper coding standards and ethical testing practices.

**Respect for Colleagues**
- Help and support team members
- Treat everyone equally
- Avoid discrimination

Example:
Mentoring junior developers without favoritism.

**Self-Development**
- Continuously learn new skills
- Keep up with modern technologies and methods

Example:
Learning cloud computing to improve software delivery.

The ACM/IEEE Code ensures software engineers create safe, ethical, and high-quality software, while protecting users, clients, and society.

## 3. Explain the main characteristics of 'professionalism'?

Professionalism means behaving in a responsible, skilled, and ethical way in your job. It is not only about doing work, but also about how a person behaves, communicates, and handles responsibilities in a professional environment.

A professional person follows rules, respects others, and always tries to improve their skills. Professionalism is very important for career success and building a good reputation.

**1. Knowledge and Skills**
A professional must have proper education and skills related to their field.
- Understand their work clearly
- Perform tasks efficiently
- Use modern tools and techniques

Without knowledge and skills, a person cannot perform work properly or solve problems.

**2. Responsibility**
A professional must take responsibility for their work.
- Complete tasks on time
- Accept mistakes and fix them
- Be reliable and dependable

Responsible people are trusted by employers and clients.

**3. Honesty and Integrity**
Honesty means telling the truth, and integrity means doing the right thing.
- Do not cheat or lie
- Be fair in work
- Maintain trust

Honesty builds strong relationships and a good reputation.

### 4. Good Communication
Communication is very important in professional life.
- Speak clearly and politely
- Listen carefully to others
- Share ideas effectively

Good communication reduces misunderstandings and improves teamwork.

### 5. Respect for Others
A professional respects everyone in the workplace.
- Treat colleagues politely
- Value different opinions
- Avoid discrimination

Respect creates a positive and friendly work environment.

### 6. Discipline and Time Management
A professional follows rules and manages time properly.
- Be punctual
- Meet deadlines
- Follow company policies

Discipline helps in completing work efficiently and maintaining order.

### 7. Professional Appearance and Behavior
A professional should have a proper appearance and behavior.
- Dress properly
- Behave politely
- Maintain confidence

Good appearance creates a positive impression.

### 8. Teamwork
A professional should be able to work with others.
- Cooperate with team members
- Help others
- Work towards common goals

Teamwork increases productivity and success.

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